3. Find the vein. Once the tourniquet is in place, you will ***palpate*** to find the vein by pressing down on the patient's skin with the tip of your index finger. Bounce your finger without leaving the surface of the skin, starting in the middle of the antecubital region until you feel something in the arm. The vein will feel spongy and bouncy. Instructing the patient to clench their fist can help the veins become more prominent. Be sure not to have the patient vigorously open and close their hand, but to clench and hold, as "pumping" the hand can cause elevations in the potassium concentration of the specimen. Once you have found a vein, run your finger a short distance up the vein to find the direction that it lies in the region. Pay attention during palpation to determine the size and depth of the vein in the patient's arm. If you feel something that feels firmer or chordlike, you are palpating a tendon. These should be avoided during blood collection to prevent injury to your patient. You will know if you have accidentally found an artery because if you pause your finger there for a second, you will feel a pulse. Arteries are not used for venipuncture and should be avoided.
4. Choose the best needle. Select the appropriate needle size. When the vein feels difficult to access, use a butterfly needle. Avoid using a 25-guage needle for adult venipunctures, as too small of a needle can hemolyze the blood, leading to inaccurate test results.
5. Assemble the needle and tube holder. Set up your needle by removing the cap, revealing a gray rubber sheath that covers the needle that inserts into the vacuum tube. The butterfly attaches similarly by screwing the gray sheath-covered needle into the smaller of the two openings in the tube holder.
6. Cleanse the site. Once the best vein has been chosen, cleanse the site with the alcohol pad using a back-and-forth rubbing motion, scrubbing the skin in an area 2 to 3 inches in diameter around the location where you will insert your needle. Allow the site to dry for 30 seconds to 1 minute to ensure it is properly cleansed and prevent burning when the needle is inserted. Do not blow or fan the area to encourage drying, as this can cause contamination.
7. Uncap the needle. After the area has been cleansed, uncap the needle. Look at the needle to confirm that there are no imperfections.
8. Hold the tube holder in your dominant hand with two fingers underneath and your thumb on top of the adapter. Place the needle bevel up and position at 15° to 30° for insertion. Choose 15° for a shallow vein, while a deeper vein would require an angle closer to 30°. Refer to the video to help you determine the proper angle for insertion.

palpate. To inspect by feel or touch for a vein to determine the depth, dimension, and direction.

The Three Ds of Finding a Good Vein

- Dimension: Palpating for the size of the vein helps you choose the best needle.
- Depth: Feeling how deep the vein is in the arm determines the angle of the needle.
- Direction: Discovering the direction of the vein in the body helps guide the direction of the needle, helping the needle follow the direction that the vein runs in the body.

Needle Position

